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Clinical management of combined tuberculosis and diabetes

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SUMMARY

Optimal management of combined tuberculosis (TB) and diabetes (DM) is important but challenging in terms of achieving good disease outcomes and avoiding toxicity, drug interactions and other challenges. DM management during anti-tuberculosis treatment, aimed at improving TB treatment outcomes and reducing DM-related morbidity and mortality, consists of glycaemic control and measures to reduce the risk of cardiovascular disease. Metformin, the glucose-lowering drug of choice for TB patients, has no meaningful interaction with rifampicin (RMP), and may reduce TB mortality. Insulin is effective for severe hyperglycaemia, but has several disadvantages that limit its use in TB patients. Cardiovascular risk assessment should be considered in TB-DM patients to guide management in terms of counselling and prescription of antihypertensive, lipid-lowering and anti-platelet treatment. With regard to anti-tuberculosis treatment,

DM is associated with an increased risk of drug resistance, lower exposure to anti-tuberculosis drugs, treatment failure and recurrent TB. Patients therefore need careful assessment before, during and possibly after anti-tuberculosis treatment. Although no studies have been performed, anti-tuberculosis treatment may also have to be prolonged or intensified in terms of regimen or drug dosage if DM is present. With regard to service delivery, combined treatment should probably be administered, supervised and monitored as much as possible in a TB clinic. Local circumstances and severity of DM will guide the need for referral of patients to specialised DM care, and continuation of DM care after completion of anti-tuberculosis treatment. More data are also needed for the management of TB-DM patients with human immunodeficiency virus co-infection.

KEY WORDS: DM; TB; treatment; metformin

THE GLOBAL EPIDEMIC of type 2 diabetes mellitus (DM), which affected an estimated 425 million individuals in 2017 and is estimated to increase to 629 million in 2045, has enormous

medical and social consequences.¹ More than 80% of type 2 DM is found in low- and middle-income countries. It is often undiagnosed and is complicated by cardiovascular complications, and eye, foot and kidney problems. DM also increases the risk of many infections and their complications.² With regard to TB, DM increases the risk of *Mycobacterium tuberculosis* infection and active TB, as well as the risk of drug resistance, treatment failure and recurrent disease. Poorly controlled DM has the strongest effects on TB susceptibility and unfavourable TB outcomes, and although evidence is still limited, as shown in a recent systematic review,³ there is growing evidence that better DM control may reverse some of

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these risks.^{4–6} In patients with dual disease, there is also evidence that poor lifestyle practices (for example, the continuation of cigarette smoking) can significantly increase the risk of death. It is clear therefore that the clinical management of combined TB and DM is important.

The scientific literature largely focuses on DM prevalence among TB patients, with relatively little evidence to guide clinicians in the clinical management of combined TB and DM. This is despite the fact that management may not necessarily be the same when treatment for the two diseases is combined. For example, anti-tuberculosis treatment or its monitoring may have to be intensified, as TB treatment outcomes are worse among TB patients with DM. Similarly, glycaemic control and other DM-related treatment may have to be adjusted because of TB-associated inflammation, drug interactions and other factors.

This review will discuss some of the challenges and considerations related to combined treatment of TB and DM, providing practical recommendations where possible. We will focus separately on DM management and anti-tuberculosis treatment in TB-DM, making brief reference to the combination of TB-DM-HIV (human immunodeficiency virus), and on challenges and possible solutions with regard to health services delivery. Screening of TB patients for DM and vice versa,⁷ as well as the underlying biology,⁸ and aspects related to integration of services for TB and DM are extensively reviewed in other articles of this State of the Art series, and are not included in this paper. We used a previous extensive narrative review on this topic,⁹ more recent relevant scientific literature, unpublished observations and data from the TANDEM project,¹⁰ and practical recommendations, as included in an upcoming guide issued by the International Union Against Tuberculosis and Lung Disease (Paris, France) and the World Diabetes Foundation (Brussels, Belgium),¹¹ specifically targeted at lower-resource settings, which carry the heaviest burden of both DM and TB.

DIABETES MANAGEMENT IN PATIENTS WITH ACTIVE TUBERCULOSIS

The management of DM is aimed at reducing short-term and long-term complications such as cardiovascular disease, eye and kidney problems and foot amputations. DM management consists of life style counselling (diet, weight loss, physical activity, smoking cessation, avoiding excess alcohol); treatment with blood glucose-lowering drugs, measures to reduce the risk of cardiovascular disease and associated complications that include antihypertensive medications, lipid-lowering drugs and anti-platelet drugs if indicated; and management of specific complications such as diabetic foot and eye problems.

For patients with combined DM and active TB, the priority is to successfully treat TB. Having said that, severe hyperglycaemia, which can be symptomatic and is likely to affect TB outcomes, should be avoided. Furthermore, not only DM, but also TB itself, is associated with an increased risk of cardiovascular complications such as myocardial infarction^{12,13} and stroke,¹⁴ and this may be an explanation for the higher rate of deaths in the first few months of anti-tuberculosis treatment in those patients who also have DM.^{15,16}

It is important to note that TB-DM patients form a heterogeneous group, consisting of those with 'known' (previously diagnosed) and 'new' DM, with hyperglycaemia ranging from mild to severe, variable duration of disease and with highly variable comorbidity, disease complications and treatment needs. In the TANDEM cohort, around 74% of TB-DM patients had previously diagnosed DM, while 26% were newly detected as a result of DM screening.¹⁰ A substantial proportion of patients with 'known' DM were no longer under DM care when their TB was diagnosed, had poor adherence to DM medication and some had never even been started on DM treatment (TANDEM consortium, unpublished). TB-DM patients also differ in terms of DM complications and cardiovascular or other comorbidities. The heterogeneity of DM in TB patients is particularly notable between different countries, most likely due to genetic, environmental, nutritional and behavioural factors, and differences in the accessibility and quality of health services. It is evident that this heterogeneity has important implications for DM management.

Glycaemic control

Treatment targets

An accepted target for glucose control in DM is glycated haemoglobin (HbA1c) < 7% (53 mmol/mol), as recommended by the American Diabetes Association,¹⁷ although the American College of Physicians recently recommended HbA1c levels of between 7% and 8%.¹⁸ Nevertheless, this may be hard to achieve during anti-tuberculosis treatment, particularly in the first 2 months, when active TB may aggravate hyperglycaemia, and in the presence of severe or longstanding DM, drug interactions with RMP, and altered patterns of food intake and energy expenditure during TB disease and treatment recovery.⁹ Moreover, the benefits, potential risks and costs of additional efforts to achieve more stringent glucose control and reach the accepted target are currently unknown.³ It is also important to realise that TB-associated inflammation can lead to temporary hyperglycaemia, which often spontaneously improves with anti-tuberculosis treatment.¹⁹ As a general rule, dysglycaemia in patients with 'known' DM is much more difficult to revert than in 'new' DM (TANDEM,

Table 1 Common glucose-lowering drugs used for managing diabetes mellitus in tuberculosis patients

Characteristic	Metformin	Sulphonyl urea derivatives	Insulin
Drug of choice	First choice	Alternative oral drug	Use if HbA1c > 10% or symptomatic hyperglycaemia
Risk of hypoglycaemia	No	Yes	Yes
Starting dose	500 mg od or bid, titrated to a maximum dose of 2000 mg daily	Gliclazide 40–80 mg od Glibenclamide 2.5–5 mg od Glimepiride 1–2 mg od Glipizide 5 mg od	10 units basal insulin per day
Interaction with RMP	Not clinically relevant	Yes, 30–80% lower efficacy with RMP	None
Main side effects/toxicity	Gastro-intestinal Lactic acidosis (rare)	Hypoglycaemia	Hypoglycaemia
Use in reduced kidney function	Dose adjustment if eGFR < 45 ml/min; contraindication if eGFR < 30 ml/min	Increased risk of hypoglycaemia Preference gliclazide	Can be safely used
Cardiovascular events	Recognised benefit	Neutral	Neutral

HbA1c = glycated haemoglobin; od = once a day; bid = twice a day; RMP = rifampicin; eGFR = estimated glomerular filtration rate.

unpublished data). Although this is not evidence-based, a more realistic treatment target—particularly under programmatic conditions—may therefore be HbA1c < 8% and a target of random blood glucose/fasting blood glucose < 11.1 mmol/l (200 mg/dl) during the treatment of TB disease, which is in line with those for DM management in persons with significant comorbidity.¹⁷

Choice of glucose-lowering drugs

The next question is how optimal glycaemic control can be achieved. To date, there is no guiding evidence as to what drugs should be used, how treatment should be monitored, who should best deliver DM treatment, and how this should be adjusted for patients with new or previously known DM, and for those with mild or severe disease. We will first address the choice of glucose-lowering drugs, discussing three classes of drugs: biguanides, sulphonyl urea derivatives and insulin. Although other drug classes are also used to treat DM, including thiazolidinediones, dipeptidyl peptidase 4 inhibitors, sodium/glucose cotransporter 2 inhibitors and glucose-dependent insulinotropic peptide receptor agonists, these medicines are more costly and there is limited evidence of their greater effectiveness.

Metformin is the first choice glucose-lowering agent recommended in type 2 DM, and there is no reason why this should be different for patients with active TB disease. The advantages of metformin include extensive experience with the drug, extremely low risk of hypoglycaemia, effectiveness, low cost, beneficial effects on cardiovascular disease,²⁰ lack of clinically relevant interaction with RMP (TANDEM, unpublished) and a potential benefit on TB itself.²¹ In a recent retrospective analysis from Taiwan, those with DM (30%) had a 1.9-fold higher mortality, but among this group metformin use was associated with lower mortality (hazard ratio [HR] 0.56, 95% confidence interval [CI] 0.39–0.82).²² Its two main

disadvantages are gastro-intestinal side effects, which may be worse when taken together with anti-tuberculosis drugs (TANDEM, unpublished), and increased toxicity, including the development of lactic acidosis in patients with decreased kidney function. Lactic acidosis, usually presenting as vomiting, abdominal pain, signs of hypovolemia, followed by Kussmaul breathing, neurological signs, cardiorespiratory, kidney or liver failure and finally death, is extremely rare, but it may be fatal if unrecognised and untreated.²³ The dose of metformin needs adjustment with a renal clearance (estimated glomerular filtration rate) < 50 ml/min (Table 1).

Sulphonyl urea derivatives can be used as second choice oral glucose-lowering agents, probably as 'add-on' to metformin if metformin alone is ineffective or if there is intolerance or a contraindication to metformin. The most widely used sulphonyl urea derivatives are gliclazide, glibenclamide, glimepiride and glipizide. The two main disadvantages are the risk of hypoglycaemia and strong drug interactions with RMP that show wide inter-individual variation but result in their efficacy being reduced by 30–80%.

Some argue that insulin is the preferred glucose-lowering treatment for TB-DM patients; however, particularly under programmatic conditions, insulin is probably the third choice, except for sick and hospitalised patients or patients who were already using insulin before the TB diagnosis. Insulin is indicated in cases of severe hyperglycaemia (for example, HbA1c > 10% or blood glucose > 18 mmol/l). It has unlimited efficacy, but it is also more costly, requires refrigeration and subcutaneous injection, and is associated with a risk of hypoglycaemia. In well-resourced settings, the use of insulin is usually accompanied by the need to self-monitor blood glucose levels through glucometers. Unavailability, insecure supply or high costs of DM medication, as well as issues related to storage and use of insulin, all

compromise glycaemic control in many resource-constrained settings.^{24,25}

Adjustment of treatment according to patient characteristics and local circumstances

Disease phenotype and severity as regards both TB and DM and local circumstances will guide the choice, timing and dosing of glucose-lowering drugs. Patients who are under DM treatment at the time of TB diagnosis can usually continue their medication, although worsening hyperglycaemia as a result of TB may require intensification (e.g., increased insulin dose). Oral glucose-lowering drugs may have to be substituted or adjusted, as except for metformin their metabolism is increased and therefore their glucose-lowering effect is reduced with concurrent use of RMP.

In TB patients with newly diagnosed or known yet untreated DM, the choice and timing of glucose-lowering drugs depends on the level of hyperglycaemia and local circumstances. If the hyperglycaemia is mild (e.g., HbA1c < 8%), initiation of glucose-lowering drugs can probably be postponed for at least 2–8 weeks. Mild hyperglycaemia often disappears with anti-tuberculosis treatment only, and early start of glucose-lowering treatment may jeopardise successful TB treatment due to side effects, drug toxicity, the pill burden or the difficulty for practitioners and patients to focus on two diseases at the same time.

Most TB patients are managed in ambulatory care and monitored at weekly or wider intervals in community or hospital out-patient clinics. In the early phase of anti-tuberculosis treatment, TB patients should preferably not be referred to specialised DM services because of the risk of transmission of *M. tuberculosis* to those working in or attending these clinics. Separate management of TB and DM also has the risk of unrecognised drug interactions, medication errors and reduced retention in either TB or DM treatment. It is therefore preferable for DM treatment to be delivered at the TB clinic, as reported previously.²⁶

One question is how DM can best be managed in TB clinics. To achieve optimal glycaemic control during TB treatment, monitoring of blood glucose during the course of TB treatment may have to be more frequent. However, frequent monitoring is associated with additional costs, and tools and skills for glucose monitoring and DM treatment may be lacking in TB or pulmonary clinics. As such, a less intense schedule, preferably following the established decision points in anti-tuberculosis treatment after 2 and 6 months, would offer significant advantages. To address this dilemma, the TANDEM project has conducted a randomised trial in Indonesia to evaluate the effect of regular scheduled glucose monitoring and algorithm-driven adjustment of DM medication on glycaemic control of DM in TB patients

(NCT02106039). Standard algorithms were used for patients with known or newly diagnosed DM and different levels of hyperglycaemia. The study has been completed and is currently being analysed.

Cardiovascular risk assessment and management

Atherosclerotic cardiovascular disease, including myocardial infarction, stroke and peripheral arterial disease, is the leading cause of morbidity and mortality for individuals with DM. The higher reported rates of mortality in TB patients with DM may be caused by these cardiovascular complications rather than by TB itself.^{15,16} It should be noted that TB patients (even in the absence of DM) experience more cardiovascular events. Cohort studies in Taiwan have shown that patients with newly diagnosed pulmonary TB have a 40% increased risk of acute coronary syndrome¹³ and a 50% higher risk of ischaemic stroke¹⁴ than controls without TB. Similarly, an analysis of insurance claims in the United States has shown that acute myocardial infarctions were two-fold higher among individuals with a history of active TB.¹²

Cardiovascular risk assessment is focused on four possible interventions: life style counselling, antihypertensive treatment, lipid-lowering treatment (statins) and treatment to reduce platelet aggregation (aspirin). To our knowledge, there is no published evidence or experience with respect to cardiovascular risk assessment and management in TB patients with newly diagnosed DM. We suggest a number of targets, related interventions and specific considerations for combined active TB and DM in Table 2.

Successful initiation of anti-tuberculosis treatment is of much greater importance in patients with newly diagnosed TB and DM than assessment and management of cardiovascular risk. We therefore suggest that the only considerations would be initiation of aspirin for those with established cardiovascular disease (e.g., a history of stroke or myocardial infarction), and counselling for smoking cessation and reduction of alcohol consumption. After completion of the initial intensive phase of anti-tuberculosis treatment (usually at 8 weeks), patients could be counselled about healthy life styles, and antihypertensive medication and statin treatment could be started as indicated (Table 2).

ANTI-TUBERCULOSIS TREATMENT FOR PATIENTS WITH COMBINED TUBERCULOSIS AND DIABETES MELLITUS

Currently recommended anti-tuberculosis treatment is similar for patients with combined TB and DM compared to those with TB only. However, this may have to be reconsidered, as DM is associated with anti-tuberculosis drug resistance,²⁷ slower treatment response, and higher rates of toxicity, failure and

Table 2 Cardiovascular disease risk assessment and management in patients with diabetes mellitus and active TB disease

Cardiovascular risk	Target	The intervention	Specific considerations in TB patients
Smoking	Stop smoking	Counselling	Relevant for TB treatment outcomes and reducing relapse rates of disease
Obesity	Body mass index > 23 (Asian) or > 25 (other)	Counselling (diet, physical activity)	Often ~10% weight gain as a result of anti-tuberculosis treatment
Excessive alcohol consumption	Avoid alcohol intake during anti-tuberculosis treatment	Counselling	Risk of liver dysfunction associated with anti-tuberculosis drugs
Hypertension	<130/80 mmHg	Antihypertensive treatment	RMP reduces the efficacy of some antihypertensive drugs (calcium channel blockers and ACE inhibitors) No interaction with thiazide diuretics ACE inhibitors cause cough in 10–15% of patients
Hyperlipidaemia	LDL < 2.6 mmol/l (100 mg/dl)	Statins: 1) for those aged >40 years; 2) for those with previous cardiovascular disease	Rifampicin reduces the efficacy of most statins
Established cardiovascular disease (previous myocardial infarct, stroke, peripheral arterial disease)	Secondary prophylaxis	Aspirin Statin	Risk of bleeding (e.g., haemoptysis in pulmonary TB)

TB = tuberculosis; RMP = rifampicin; ACE = angiotensin-converting-enzyme; LDL = low-density lipoprotein.

recurrent TB. First, the length of anti-tuberculosis treatment might have to be adjusted, as is common practice already in some countries, including China.²⁸ In a large retrospective cohort study in Taiwan, a 9-month treatment regimen was associated with a lower rate of recurrent TB than the 6-month regimen (HR 0.76, 95%CI 0.59–0.97).²⁹

In addition to length of treatment, higher-dose TB treatment may also help improve treatment outcomes. Some studies have reported associations between DM and lower concentrations of anti-tuberculosis drugs,^{30,31} although this may be partly explained by inadequate correction for body weight in TB patients with DM, who are generally significantly heavier than those without DM.³² In an observational study in the United States, therapeutic drug monitoring for isoniazid (INH) and RMP after 2 weeks of treatment was associated with significantly shorter time to sputum culture conversion among patients with combined TB and DM (42 vs. 62 days; $P = 0.01$).³³

Another factor to consider is the apparent association between DM and anti-tuberculosis drug resistance. A recent meta-analysis that included 9289 patients from 13 studies found a significant association between DM and multidrug-resistant TB (MDR-TB) (odds ratio 1.71, 95%CI 1.32–2.22).²⁷ Possible factors contributing to this association include higher rates of acquisition of drug resistance with higher initial bacterial load, slower response to treatment, or nosocomial acquisition of drug resistance with higher rates of hospital admission. Alternatively, individuals with DM may be more susceptible to drug-resistant

M. tuberculosis isolates; some drug resistance mutations lead to ‘loss of fitness’ of *M. tuberculosis*; however, such ‘less fit’ strains might still cause active TB among individuals with DM due to their decreased host immune function against TB.³⁴ Whatever the underlying cause, TB-DM patients should probably be prioritised for drug susceptibility testing (DST) in those settings where DST is not routinely performed for all patients, both at baseline and during follow-up. The Xpert® MTB/RIF assay (Cepheid Inc, Sunnyvale, CA, USA), which is being scaled up globally, allows the confirmation of *M. tuberculosis* infection and the detection of RMP resistance (sequivalent to MDR-TB) within 2 h. In 2013, the World Health Organization recommended that the assay be considered as the initial diagnostic test for all people requiring investigation for TB. While this is yet to happen for all patients, TB-DM patients should be strongly considered for Xpert testing.

Finally, anti-tuberculosis treatment may have to be monitored more closely in patients with combined DM and TB than in those with TB only, not only because of the higher risk of treatment failure, but also because of a higher risk of drug toxicity, side effects and drug-drug interactions, necessitating adjustment of drug dosage or regimen. Although there is no evidence to support this, a high pill burden and higher rates of side effects in TB-DM patients might compromise adherence, particularly if more toxic second-line drugs are used for MDR-TB.

Table 3 Special considerations when managing human immunodeficiency virus infection, diabetes and TB together

The issue	Special considerations
TB-associated IRIS	Corticosteroids that may be given for IRIS which can increase levels of hyperglycaemia, requiring more frequent assessment
Drug-drug interactions	Many ARVs can affect metabolism of glucose-lowering drugs, statins, antihypertensive drugs and anti-platelet drugs Metabolism of most ARVs is increased by rifampicin. Integrase inhibitors (in a double dose) are the preferred choice, efavirenz (standard dose) is an alternative, protease inhibitors should be avoided Dolutegravir reduces metabolism of metformin; metformin dose should be lowered by 50%

TB = tuberculosis; IRIS = immune reconstitution inflammatory syndrome; ARV = antiretroviral.

OTHER CONSIDERATIONS RELATED TO THE TREATMENT OF COMBINED DIABETES AND TUBERCULOSIS

A number of other factors need to be considered when treating patients with combined DM and TB. For example, there is a higher risk of drug-drug interactions. RMP increases the metabolism of many drugs commonly used by DM patients, including statins, all oral DM drugs except metformin, calcium channel blockers, angiotensin-converting-enzyme inhibitors, digoxin, warfarin, and some antihypertensive drugs.³⁵ In addition, TB-DM patients may have more symptoms, or symptoms that are more difficult to interpret. For example, abdominal symptoms in an elderly TB patient with DM may be a side effect of anti-tuberculosis drugs, hepatotoxicity, side effects of metformin, metformin-associated lactic acidosis, inferior myocardial infarction or bowel ischaemia. Furthermore, drug toxicity may be worsened: INH can aggravate diabetic neuropathy, and TB-DM patients have a higher risk of liver and kidney toxicity. This is also relevant for the management of MDR-TB: patients with DM probably have a higher risk of renal toxicity with aminoglycosides, and a higher risk of neuropathy with linezolid. Furthermore, a high pill load when patients are treated for both diseases may lead to missed doses, incorrect drug intake, treatment interruptions or loss to follow-up. These and other issues necessitate more careful assessment before and during combined TB and DM treatment.

What if patients with tuberculosis and diabetes also have human immunodeficiency virus infection?

In 2016, there were an estimated 1.03 million people worldwide with HIV-associated TB, and this was associated with an estimated 370 000 HIV-related TB deaths.³⁶ Although the number of new HIV infections worldwide is levelling off, some TB-endemic countries, such as the former Soviet republics and Indonesia, are experiencing significant growth of the HIV epidemic. HIV is associated with DM; in a study in Tanzania, glucose metabolism disorders were six-fold more common among HIV-infected individuals than age- and weight-based controls.³⁷ In Ethiopia and South Africa, the longer duration of

HIV treatment was associated with increased incidence of DM.^{38,39} Finally, in part due to the successful roll-out of antiretroviral treatment, increasing numbers of HIV-infected patients get older and become at increased risk of developing DM.⁴⁰ A growing number of people may therefore be affected by DM, HIV and active TB at the same time. In such cases, there is an even higher risk of drug-drug interactions, toxicity and overlapping side effects. Some specific considerations for treating combined HIV, DM and TB are given in Table 3.

CONCLUSION

There are many challenges associated with the combined management of TB and DM and TB, DM and HIV. To date, most studies have been epidemiological, examining DM prevalence among TB patients or comparing disease characteristics and outcomes of TB patients with and without DM. Many questions related to optimal management therefore remain unanswered.⁴¹ How much will DM patients benefit from optimising glycaemic control and cardiovascular risk management, and how can this be achieved? Should anti-tuberculosis treatment be adjusted in terms of duration, drug regimen, dosage or monitoring to reduce treatment failure, TB recurrence and drug toxicity? And what should be done if the patient is also HIV-infected? Besides clinical management, we also have to define what is needed in terms of optimal health service delivery. In terms of infection risks, it seems logical to administer, supervise and monitor combined treatment as much as possible in a TB clinic, but local circumstances and the severity of DM will guide the need for referral of patients to specialised DM care, and the best ways to continue DM care after completion of anti-tuberculosis treatment. These and other issues clearly require more study, including randomised clinical trials.

Conflicts of interest: none declared.

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R É S U M É

Une prise en charge optimale de l'association tuberculose (TB) et diabète (DM) est importante mais délicate, en termes d'obtention des bons résultats du traitement tout en évitant la toxicité, les interactions médicamenteuses et d'autres défis. La prise en charge du DM pendant le traitement antituberculeux vise à améliorer les résultats du traitement de la TB et à réduire la morbidité et la mortalité liées au DM ; elle consiste en le contrôle de la glycémie et en mesures visant à réduire le risque de problèmes cardiovasculaires. La metformine, le médicament hypoglycémiant de premier choix pour les patients TB, n'a pas d'interaction significative avec la rifampicine (RMP) et peut réduire la mortalité due à la TB. L'insuline est efficace pour les hyperglycémies majeures, mais a plusieurs inconvénients qui limitent son utilisation chez les patients TB. L'évaluation du risque cardiovasculaire doit être envisagée chez les patients TB-DM afin de guider la prise en charge en termes de conseil et de prescription de traitement anti-hypertenseur, réducteur des lipides et

anti-agrégant plaquettaire. En ce qui concerne le traitement de la TB, le DM est associé à un risque accru de pharmacorésistance, de moindre exposition aux médicaments de la TB, d'échec du traitement et de rechute de la TB. C'est pourquoi les patients ont besoin d'une évaluation soignée avant, pendant et peut-être après le traitement de TB. Bien qu'aucune étude n'ait été réalisée, le traitement de la TB pourrait également devoir être prolongé ou intensifié en termes de protocole ou de posologie des médicaments en présence de DM. En ce qui concerne la prestation de services, le traitement combiné devrait probablement être administré, supervisé et suivi dans la mesure du possible dans un centre de TB. Les conditions locales et la gravité du DM vont guider le besoin de référence des patients vers un centre spécialisé dans la prise en charge du DM et la poursuite des soins du DM après achèvement du traitement de TB. Davantage de données sont également nécessaires pour la prise en charge des patients TB-DM ayant une co-infection au virus de l'immunodéficience humaine.

R E S U M E N

Un tratamiento combinado óptimo de la tuberculosis (TB) y la diabetes (DM) es importante, pero plantea problemas con respecto a lograr desenlaces clínicos favorables y evitar la toxicidad, las interacciones medicamentosas y otras dificultades. El manejo de la DM durante el tratamiento antituberculoso, encaminado a mejorar los desenlaces terapéuticos de la TB y disminuir la morbilidad y la mortalidad causadas por la DM, consiste en regular la glucemia y adoptar medidas que disminuyan el riesgo de aparición de una enfermedad cardiovascular. La metformina es el hipoglucemiante de primera opción en los pacientes con TB, no presenta interacciones importantes con la rifampicina y disminuye la mortalidad por TB. La insulina es eficaz en los casos de hiperglucemia grave, pero presenta varias desventajas que limitan su utilización en los pacientes con TB. Se debe considerar la evaluación del riesgo cardiovascular en los pacientes con TB y DM, con el fin de orientar el manejo en materia de asesoramiento y prescripción de tratamiento antihipertensivo, hipolipidemiante y antiagregante plaquetario. Con respecto al tratamiento

antituberculoso, la DM se asocia con un mayor riesgo de farmacorresistencia, menor exposición a los fármacos antituberculosos, fracaso terapéutico y recaída de la enfermedad. Por lo tanto, los pacientes precisan una evaluación cuidadosa antes de iniciar el tratamiento antituberculoso, durante el mismo y tal vez después de haberlo terminado. Aunque no se han realizado estudios específicos, también puede ser necesario prolongar el tratamiento de la TB o intensificar el esquema o la dosificación cuando existe DM concomitante. En cuanto a la prestación del servicio, se recomienda administrar un tratamiento combinado, supervisado y monitorizado en un consultorio de TB, en la medida de lo posible. Las circunstancias locales y la gravedad de la DM deben orientar la necesidad de remitir los pacientes a un servicio de atención especializada de la DM y su continuación después de haber terminado el tratamiento antituberculoso. También se precisan más datos sobre el manejo de los pacientes con TB y DM que presentan coinfección por el virus de la inmunodeficiencia humana.